

6. Implement a basic N-gram model for text generation. For example, generate text using a bigram model using python.

```
[2]
✓ 0s  import random

text = "I love natural language processing and I love python programming"

words = text.lower().split()

# Create bigram dictionary
bigrams = {}

for i in range(len(words) - 1):
    key = words[i]
    value = words[i + 1]
    if key not in bigrams:
        bigrams[key] = []
    bigrams[key].append(value)

# Generate text
word = random.choice(words)
result = [word]

for _ in range(10):
    if word in bigrams:
        word = random.choice(bigrams[word])
        result.append(word)
    else:
        break

print("Generated Text:")
print(" ".join(result))
```

```
... Generated Text:
and i love natural language processing and i love natural language
```